



# Predicting Cursor Tracking Error from Behavioral Signals for an Adaptive Decoder EMG Interface

Ishika Kanakath, Amber Chou, Sam Burden

Website:  
<https://ishikakkanakath.gitlab.io/>

ELECTRICAL & COMPUTER  
ENGINEERING

UNIVERSITY of WASHINGTON

## Introduction

Myoelectric decoders translate EMG activities into cursor or robotic control but still provide unreliable control in real world usage.

Closed-loop decoder adaptation that updates in response to the users improves performance.

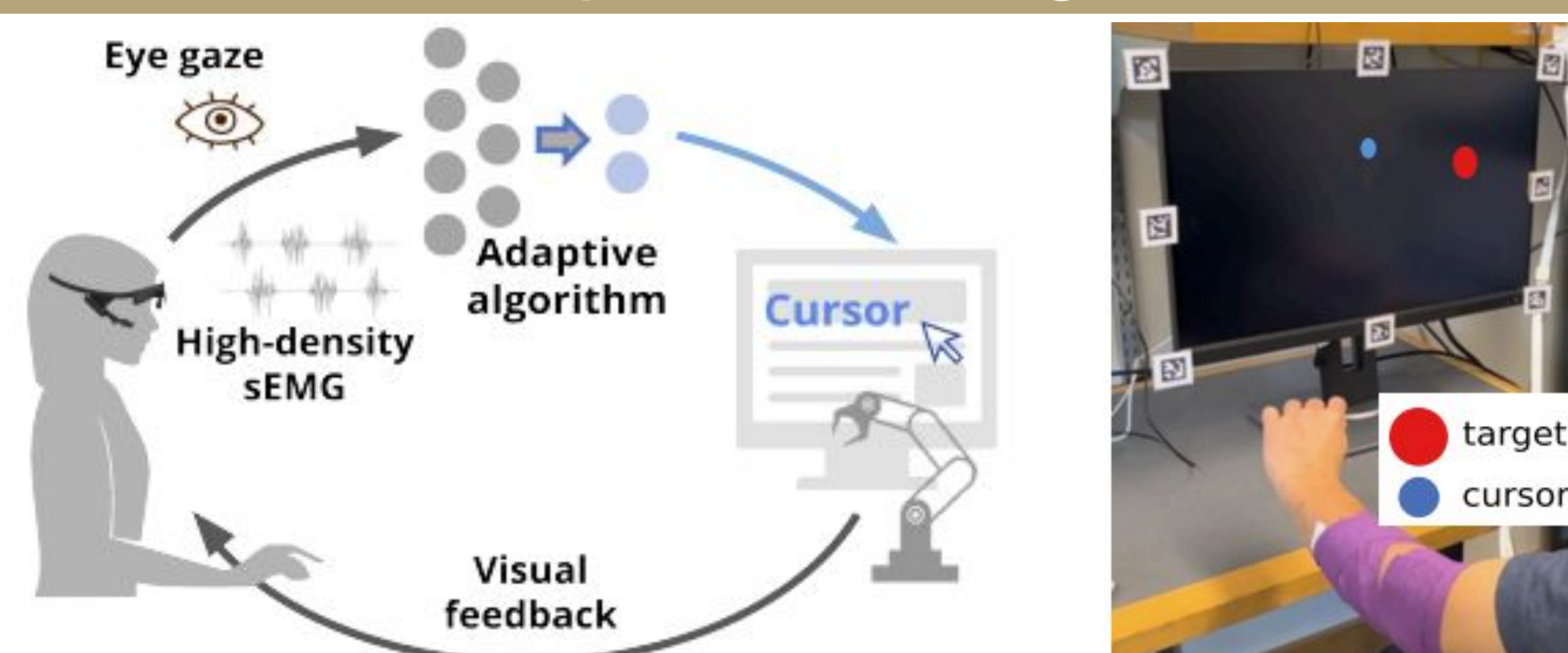
**THE GAP** : However, when and how much to update the decoder is still an open question.

**HYPOTHESIS**: Detecting periods of degraded control performance may inform when decoder updates should occur.

### Objective

We aim to evaluate how behavioral signals, derived from gaze and task data, inform closed-loop decoder performance, and thus can be used as biomarkers for future decoder adaptation.

## Experimental Design



### Closed-Loop EMG Cursor Control

Closed-loop EMG Cursor Control:

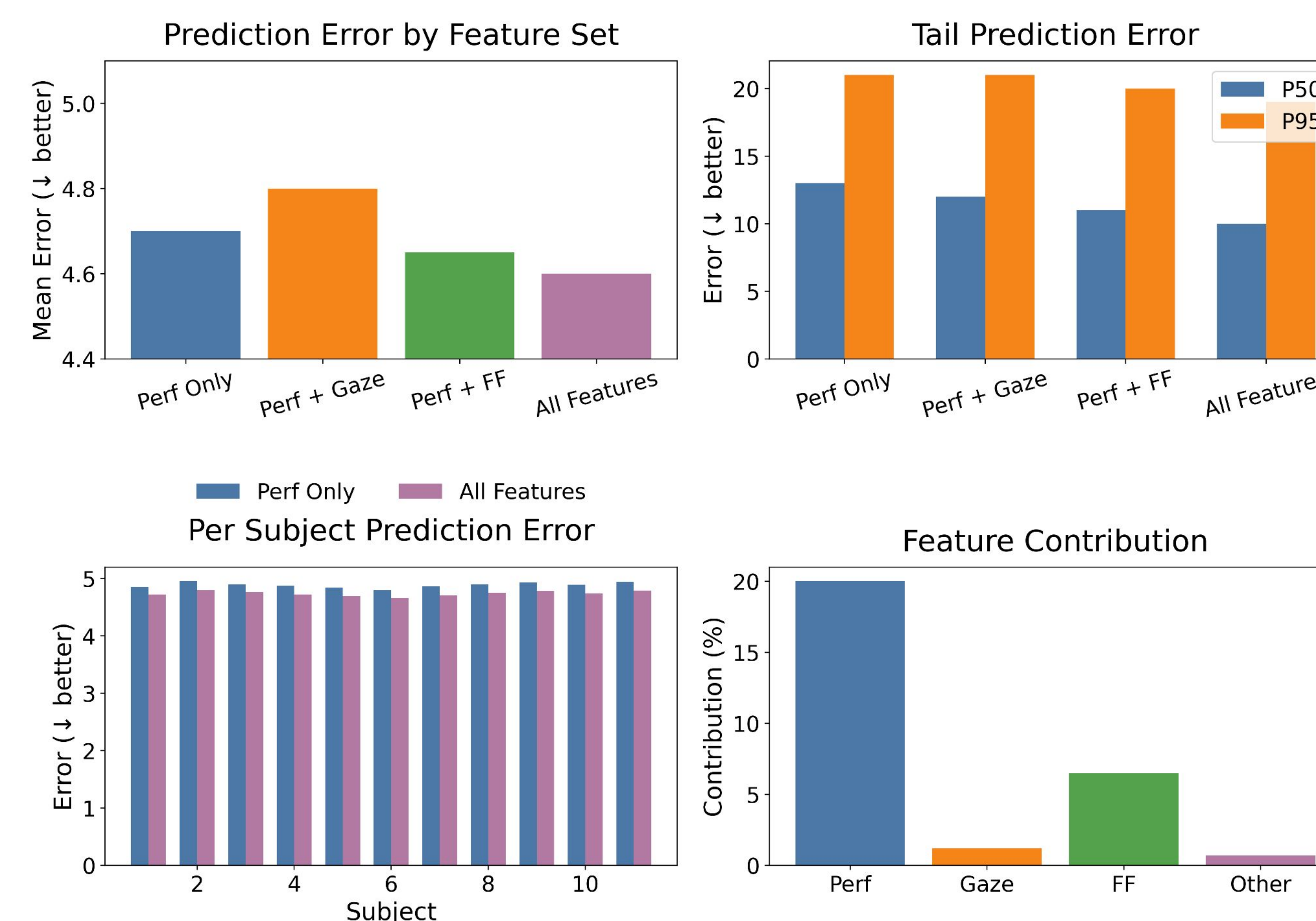
Secondary analysis from an experiment where N= 11 subjects used EMG to control cursor velocity to track continuous target. Gaze provides behavioral context about the user's interaction with the task.

Behavioral Features: (1) cursor tracking error, (2) gaze-cursor distance, (3) user's forward-model, are used to estimate upcoming control performance.

**Predictor**: Ridge+LOSO model predicts upcoming tracking error from **performance, gaze, and forward-model features**.

## Predicting Upcoming Tracking Error from Behavioral Signals

### Feature Comparison and Ablation Results



Ridge regression predicts upcoming tracking error:

$$\hat{r}(t + \Delta) = w^T x(t)$$

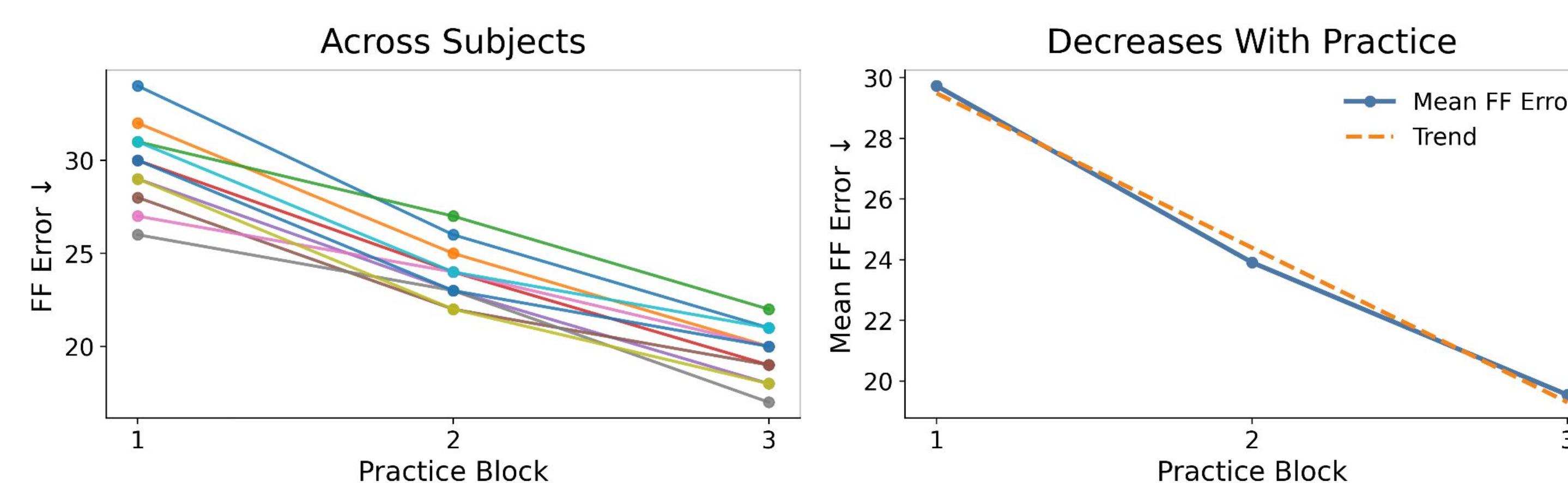
$x(t)$  :behavioral signals (performance, gaze, forward-model indicators)

$\hat{r}(t + \Delta)$  :predicted error in the upcoming window.

Predicted error identifies **periods of degraded control**, which could guide **when decoder updates occur**.

## Forward Model Learning during Coadaptation

### Forward-Model Adaptation

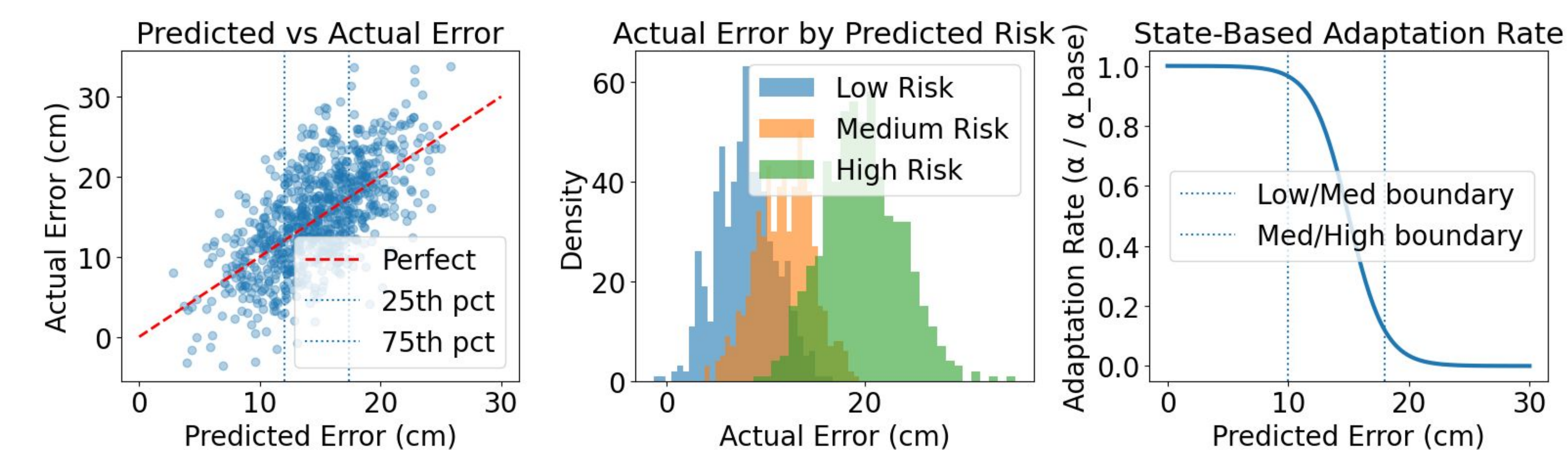


The **forward model** represents the user's internal prediction of cursor motion from muscle activity.

**Feedforward error** measures the mismatch between predicted and actual cursor motion.

Feedforward error **decreases across practice blocks**, indicating adaptation in the user-decoder interaction.

## Using Predicted Error to Guide Decoder Updates



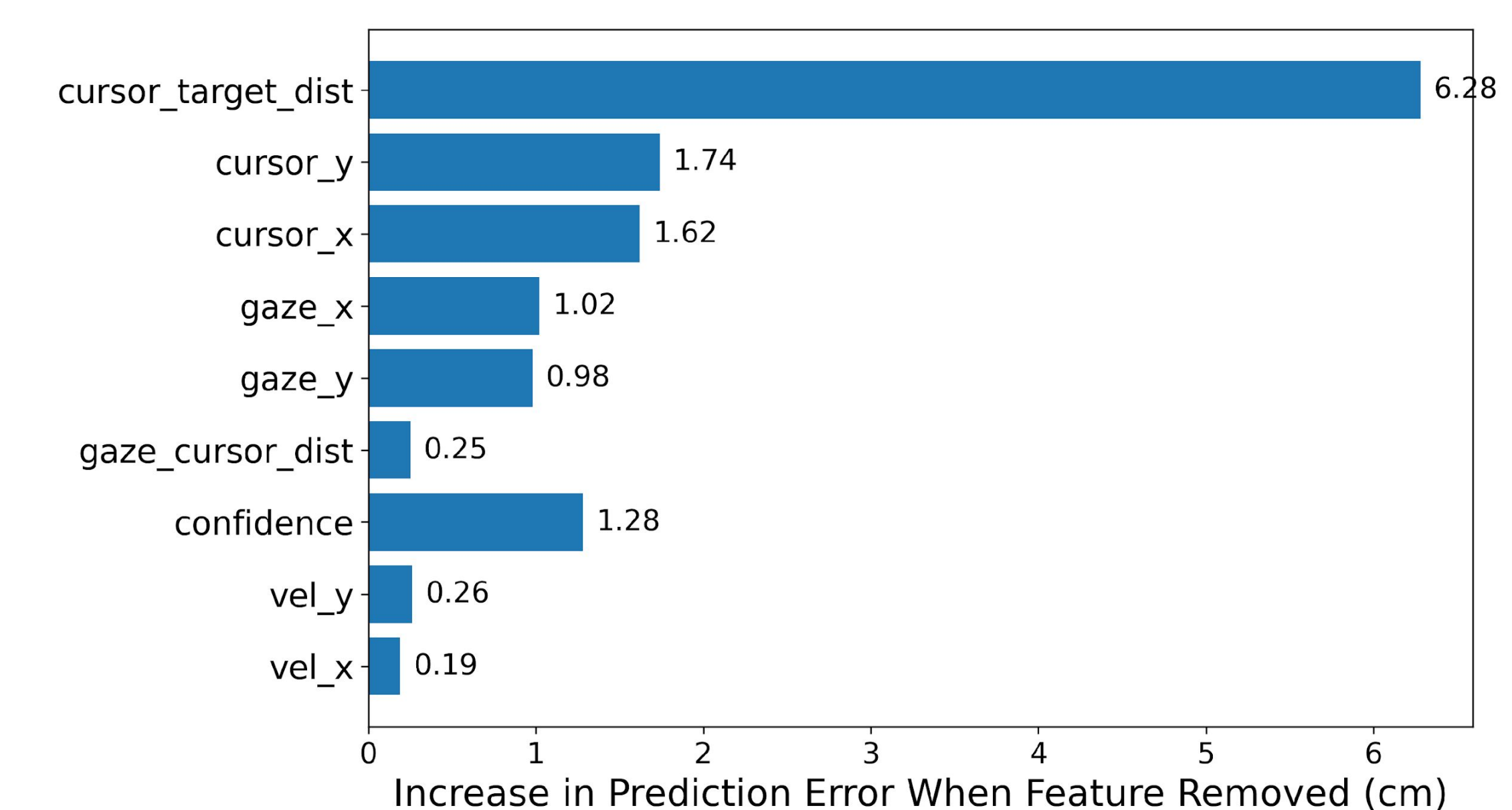
High predicted-error windows show **~2× larger P95 tracking error** than low-error windows (40.0 vs 19.1 cm).

The model predicts tracking error in the next **500-ms window**:

**Risk levels** are defined from predicted error (low, medium, high).

### Results

#### Feature Ablation: Error Increase When Feature Is Removed



Forward-model indicators provide the largest improvement in prediction accuracy, indicating that user adaptation signals strongly predict upcoming cursor tracking error.

### Takeaways

- Behavioral models, especially forward models, predict future tracking error during EMG cursor control.
- These Predictors can inform user dependent decoder adaptation, such as constantly changing adaptation rates for EMG interfaces

### Future work

- Validate the predictor in a real-time closed-loop experiment where predicted error triggers adaptive decoder updates.

## References

- [1] Chou, A. H.Y., Madduri, M., Li, S. J., Isa, J., Christensen, A., Hutchison, F. L., Burden, S. A., & Orsborn, A. L. (2024). Using Eye Gaze to Train an Adaptive Myoelectric Interface. bioRxiv. doi: 10.1101/2024.04.08.588608
- [2] Madduri, M. M., Yamagami, M., Li, S., Burckhardt, S., Burden, S. A., & Orsborn, A. L. (2025). Computational Framework to Predict and Shape Human-Machine Interactions in Closed-loop, Co-adaptive Neural Interfaces. bioRxiv. doi: 10.1101/2024.05.23.595598